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Review

Smart batteries for powering the future

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SUMMARY

The advent of the smart era drives the groundbreaking development of smart devices, which has increased the requirement for the application of energy supply equipment. Although lithium-ion batteries are available for commercial applications, their electrochemical properties and adaptability are still limited by the intrinsic material defects and complex technological innovations. Facing the fourth industrial revolution represented by information technology and artificial intelligence (AI), novel batteries with superior electrochemical properties and outstanding reliability can be constructed based on these disruptive materials and technologies. Importantly, the intelligence concept has been introduced into the design and management of batteries. In this review, we explicitly define and discuss the meaning of "smart batteries" and categorize them into three generations based on the intelligent features of their functional characteristics. Meanwhile, the action mechanisms and application principles of smart batteries have been elaborated to provide a comprehensive understanding. Moreover, we propose feasible opinions to the problems faced in the practical development of smart batteries to promote the sustainability of batteries.

INTRODUCTION

The 2019 Nobel Prize in Chemistry was awarded to M. Stanley Whittingham, John B. Goodenough, and Akira Yoshino for their work in developing lithium-ion batteries (LIBs).[1] Since their inception, batteries have been recognized as a crucial technology for various electronics, electric vehicles, and energy storage devices. Rechargeable batteries have become essential in daily life and industrial production, serving as an indispensable energy tool.[2] Historically, energy revolutions were driven by power revolutions, with the invention of new power plants and means of transportation leading to the industrial revolution. As part of the deep integration of new generations of information technology and manufacturing, smartphones, electric vehicles, and energy storage grids have rapidly developed. The emerging evolution of intelligent devices and changing application scenarios have placed higher demands on batteries.[1,3,4]

The appearance of the first Voltaic Pile in 1799 proved that electricity can be artificially manufactured. The electrochemical performance indicators of LIBs have made great progress after 200 years of development and are now mature and commercially applicable (Figure 1A).[5] However, despite the rapid development of smart devices, battery technology has made slow progress over the past few decades, without any ground-breaking innovations. Traditional research has focused on optimizing the quality, reliability, lifetime, and safety (QRLS), but it has been limited by the difficulty of theoretically innovating electrode materials and challenges in practical applications.[6–8] It is urgent to break the function interval of the

CONTEXT & SCALE

Lithium-ion batteries are now ripe for commercial application. However, the development of rechargeable batteries has been slow in recent decades due to the intrinsic qualities of the materials and technological innovations. With the advent of the fourth industrial revolution, it is characterized by rapid advancements in high technology such as artificial intelligence. By incorporating the concept of intelligence into battery design and manufacture, the new power systems that integrate cutting-edge information technologies are poised to revolutionize the energy transformation process. Despite these advancements, the concept and understanding of smart batteries still lack clarity.

This review presents a comprehensive explanation for the definition of smart batteries. Based on the various functional characteristics and intelligence levels, smart batteries can be classified into three generations: real-time perception smart batteries, dynamic response smart batteries, and self-decision-making smart batteries. Meanwhile, the mechanisms of action and application principles of each function are discussed in detail for an in-depth understanding. Furthermore, a Visio roadmap suggesting potential issues and feasible solutions is provided to facilitate the practical development of sustainable smart batteries.

original batteries, accelerate smart battery research, and promote the development of battery technology in a new direction through the use of interdisciplinary approaches, as there is little room for developing new battery technologies based on conventional concepts.

In the 21st century, the term "fourth industrial revolution" has emerged in response to the global energy crisis and climate change. This revolution mainly focuses on artificial intelligence (AI), clean energy, biotechnology, quantum information technology, and unmanned technology, representing a shift toward the era of AI and green energy.[9] This historical energy transformation is driven by new power systems, integrating energy storage technology and modern information technologies, such as the internet, big data, and cloud computing. Many countries have formulated such plans and dedicated resources to the research and development of new battery technologies as the European Union (EU) has proposed the "Battery 2030+ Roadmap," the US has launched the "National Blueprint for Lithium Batteries 2021–2030," and China has incorporated advanced battery technology development into its Five-Year Plan for National Economic and Social Development.[10,11] Researchers have introduced the concept of intelligence into battery design and development, which involves the compatibility development of advanced technologies and materials within the context of batteries and AI.[12–14] However, a precise definition and clear understanding of what constitutes smart batteries, as well as the necessary mechanisms and technical routes to achieve deep integration, are still lacking. Thus, a clear definition of smart batteries is crucial to influence their potential as a driver for accelerating the transition to sustainable and intelligent energy storage.

With the development of AI-driven technology groups, the ability to build scientific decision-making and autonomous learning through model optimization and algorithms is evolving, in addition to traditional biological intelligence of perceiving changes to the surrounding environment or internal state and giving appropriate feedback and making autonomous decisions. Similar to the functioning of the brain, smart batteries can also perceive stimulations from the internal and external environments and generate sensory signals that transfer through the nervous system to the effector. This gives a high level of feedback on the effects of the stimulus or allows for decisions based on collected perceptual information. Smart batteries require certain functions, including perceptual function, response function, and decision-making function.

The perceptual function collects and converts information from the internal and external battery environment, allowing for information transmission, processing, storage, display, recording, and control. The response function perceives battery stimulations and responds accordingly, allowing for sensitive, rapid, appropriate, and effective feedback to the battery in a specific way. Finally, the decision-making function has the capacity for self-discipline, learning, scientific prediction, and self-maintenance, making the battery capable of self-diagnosis, self-regulation, and control based on collected complex operating-state information and building a thinking system for the battery.

The design and manufacture of smart batteries are realized by the interdisciplinary integration of materials science and engineering, instrumentation science and technology, information and communication engineering, computer science and technology, electronic science and technology, and control science and engineering. The smart battery is a comprehensive system that integrates real-time perception,

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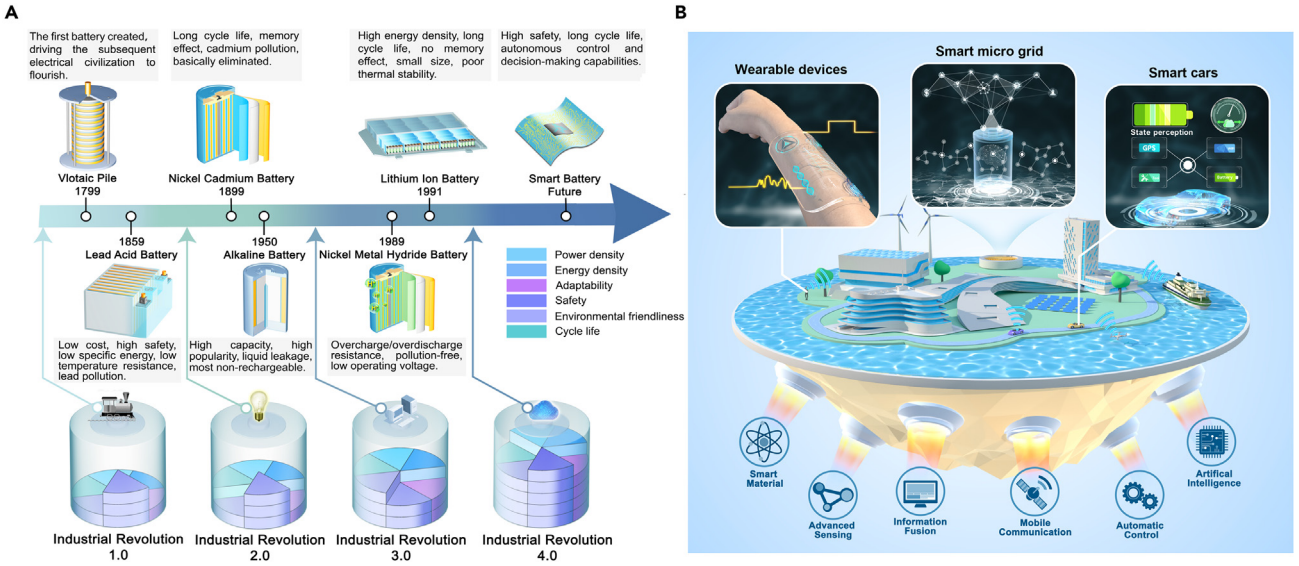


Figure 1. Battery development history and smart batteries application scenarios

(A) The evolution of battery characteristics based on the industrial revolution 1.0 to 4.0 technology and the future development of a new generation system of smart batteries.

(B) Technical support for the development of smart batteries and the demand for intelligent application scenarios. Driven by smart batteries, future wearable devices can be more flexible, adaptable, and intelligent. The safety and range of smart cars and the intelligence of other devices for batteries will be dramatically improved. Also, future energy information can be interconnected and optimally managed in urban areas.

dynamic response, and self-decision-making, as well as high-tech technologies, such as smart materials, advanced sensing, information fusion, mobile communication, automatic control, and AI. In order to improve the electrochemical performance, enhance safety and reliability, increase application adaptability, and optimize functional diversity of energy storage devices, the research on smart batteries is primarily focused on the goals of informatization, interactivity, and automation. As a key technology that promotes the rapid development of power supply equipment, a series of advanced batteries with smart characteristics will play an essential role in the fields of smart grid integration, wearable portability, electrified vehicles, intelligent equipment, and many others (Figure 1B). In this review, we categorize smart batteries into generations based on their functional characteristics and intelligent level, followed by a detailed explanation of the mechanisms and technologies of perception, response, and decision. Finally, we will provide a Visio roadmap for the challenges and opinions of smart batteries.

GENERATIONS OF SMART BATTERIES DEVELOPMENT

Smart batteries have the potential to greatly outperform the basic performance of traditional rechargeable batteries, particularly beneficial in providing additional functionality to batteries, including state sensing, self-response, and decision-making control. Sensing technology is the core support of smart batteries because it can monitor and reflect on the physical field information within the batteries. Thus, it can accurately diagnose the working state and operating environment of the batteries in real time. Additionally, smart materials enable rapid, timely, and effective responses to the battery-running state. The application of machine learning (ML) technology, in turn, elevates the directions for evaluating and predicting the battery's performance. Based on the cloud battery management system (BMS) platform's powerful computing power and storage space, a digital model of battery physical mapping to be built combines data-driven and digital twin (DT) technologies, providing

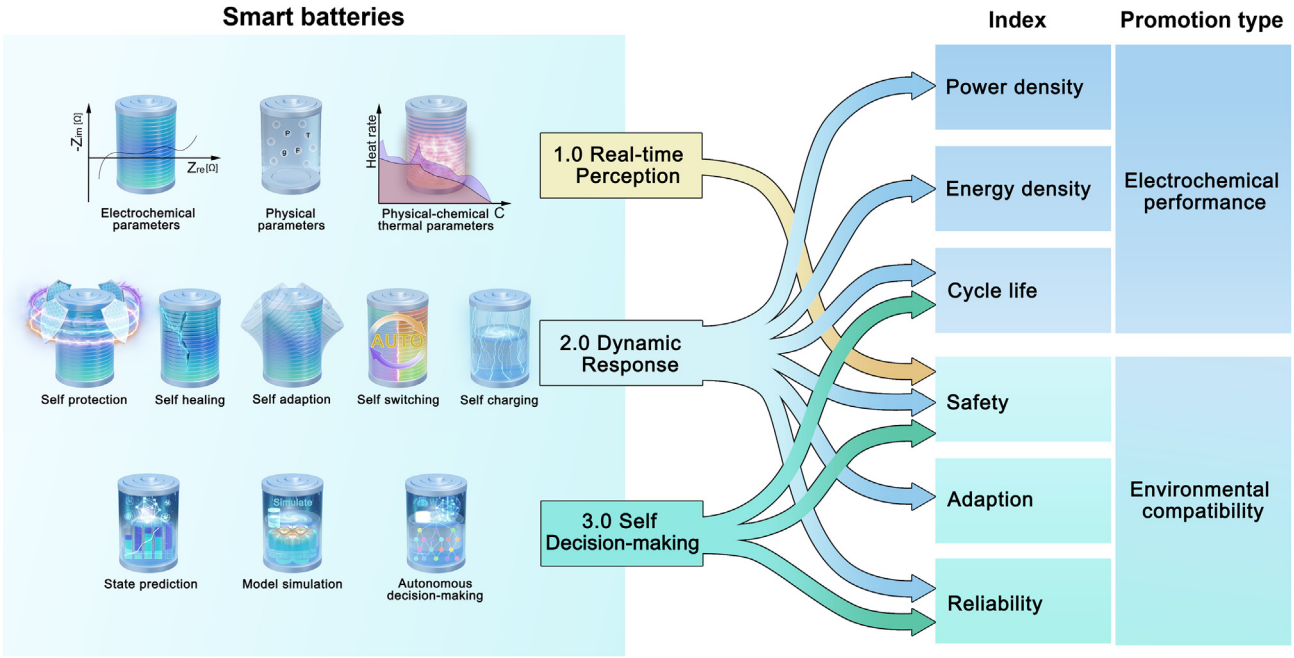


Figure 2. Generations and corresponding functions of smart batteries

Based on the intelligent features of smart functions, smart batteries are classified into three generations, real-time perception, dynamical response, and self-decision-making.

real-time monitoring and continuous learning optimization for the entire battery life cycle. While improving the data processing capability, it realizes the autonomous decision-making and control of the battery.

From the perspective of development and the intelligent features of smart functions, smart batteries can be divided into three generations, which are real-time perception smart batteries, dynamic response smart batteries, and self-decision-making smart batteries, as shown in Figure 2.

Implanting multi-dimensional physical sensors in real-time perception, smart batteries can monitor the rich physical information inside the batteries and realize the conversion of physical-chemical-digital information. These batteries accurately perceive various signals, including battery temperature, strain, decomposed gas, and pressure. This monitoring is crucial for addressing complex microscopic electrochemical reactions within the batteries, as well as detecting variations in strain and pressure. Moreover, it plays a pivotal role in mitigating safety risks such as interferences, performance degradation, and safety declines resulting from external environmental factors. This sort of battery methodically masters the information about the internal and exterior operating environments of batteries and offers more precise assessments of state of charge (SOC) and state of health (SOH). The precise monitoring of battery states enables the timely identification of potential issues, averting battery failures, and consequently elevating the overall safety of the battery system.

Dynamic response smart batteries utilize smart materials that can adapt to the changing environment. They enable the interactive process of stimulus response, which allows for timely and accurate feedback based on the state of the battery. There are several smart materials available for this purpose, including self-healing

materials, shape-memory materials, and environment-sensitive materials. Self-healing materials can repair themselves after damage, whereas shape-memory materials can recover original shape after plastic deformation. Environment-sensitive materials can perceive subtle changes in physics or chemistry and produce corresponding changes in the battery's operation. Other functional materials can perform specific functions in different environments to provide smart batteries with self-repair, self-charging, self-protection, and shape-memory features. These smart functionalities not only impact the electrochemical performance metrics of the battery, such as energy density, power density, and lifespan, but also confer better environmental compatibility in terms of safety, adaptability, and reliability.

Based on the real-time perception type and dynamic response type smart batteries, the autonomous decision-making smart batteries utilize data-driven model and DT technologies to predict and map the whole life cycle process of the batteries in a virtual space, integrating multi-discipline, multi-physical quantity, and multi-dimensional simulation processes using physical principles and sensor information acquisition and data operation. The cloud BMS is built using the historical data of the cloud, allowing for simultaneous autonomous decision-making and adjustment between the battery and the environment, battery-to-battery, and battery-to-smart management system, which provides feedback behaviors of timely online diagnosis and early warning in a controllable way, hence facilitating battery performance optimization and efficiency maximization. Furthermore, the control decisions constantly improve as the cloud data are updated, leading to enhanced accuracy of status assessment, safety warning, operational reliability, and autonomous control capacity.

Furthermore, the maturity of the three-generation smart battery technologies greatly depends on the advancements in advanced sensing, smart materials, and AI technology, which will be elaborated shortly.

SMART BATTERIES FUNCTION

Real-time perception smart batteries

Decades have witnessed the development of multiple sensing technology drives from the earliest reliance on specific gravity meters or even optical prisms to determine the SOC of lead-acid batteries.[15–17] Due to the limitations of density meters, electrochemical methods, including voltage, current, and electrochemical impedance spectroscopy (EIS), have become mainstream. EIS is a well-established nondestructive characterization technique for determining the kinetic behavior of batteries and can change over time and/or during battery operation, enabling the study of the global evolution of battery impedance related to contact and charge transfer resistance at the electrode-electrolyte interface.[18] Therefore, monitoring the impedance growth of a battery during cycling and aging can be of great help in tracking the health of the battery. With the development of electrochemical testing techniques, many traditional battery testers have been developed to measure some of the key indicators that affect the degradation of battery systems.[19] However, all these sensing techniques lead to parameter uncertainty in BMS design because of the inability to visually obtain observations inside the battery. Additionally, advanced diagnostic techniques such as X-ray, electron, neutron, energy spectrum, scanning probes, and imaging have been developed to characterize the physical properties and electrochemical performance of the battery without damage.[20–24] Although these analytical techniques facilitate the accurate quantification of chemical reactions and parameters in batteries, depending on specific equipment that is difficult

to perform on large-scale commercial batteries. Using acoustic techniques is indeed abundant in visualizing the consequences of mechanical effects, but there is an issue with the difficulty in distinguishing the exact sources.

In general, all of the above sensing techniques have different advantages to obtain key metrics for monitoring commercial battery aging under simulated operating conditions, but they are limited by gradient effects, localization effects, and are unable to provide direct access to information about the internal state of the battery.[25,26] Importantly, the acquisition of internal physical field parameters of the battery is an important basis for judging the battery operation. Due to the special characteristics of the internal battery environment, the aforementioned methods are no longer applicable, and obtaining accurate information about the internal battery is extremely challenging. To break through this limitation, disruptive innovations are injecting smart functionality into the battery mainly through the development and integration of advanced sensing technologies. Today, a rapidly evolving variety of advanced sensors have been successfully applied in various batteries to obtain combined physical-chemical parameters. The sensing information, such as temperature, strain, gas, and pressure, during the whole life cycle of the battery can be obtained by means of internal implantation, which is of great significance for further evaluation of the working state of the battery and thermal runaway warning (Figure 3A). Here, we focus on the principle of sensing application of endogenous sensors inside the battery and further elaborate the sensing mechanism with the aid of characteristic curves of each sensor action.

Temperature sensor

Battery safety has been a significant concern due to frequent safety accidents caused by thermal runaway. This phenomenon is triggered by various factors, resulting in a sharp and uncontrollable rise in battery temperature leading to battery explosion or combustion.[27,28] The current commercial BMS relies on the measurement of the battery surface temperature to prevent thermal runaway. Although implanting temperature sensors into the battery can monitor the internal temperature of the battery in real time, predict, and warn against battery safety issues. This integration of multipoint thin-film temperature sensors into the electrodes can be used to analyze the failure mechanism of Li-ion batteries and improve safety.[29] Multiple measurement points are formed on a flexible polyimide substrate using magnetron sputtering technology as shown in Figure 3BI. Thin-film sensors are attached to the electrode area with a small area of active materials removed. By measuring the relationship between the temperature and the resistance of each measurement point, the variations of the internal temperature are obtained, and a calibration curve is obtained. The structure can attenuate the effect of the sensor on capacity loss and damage to the active material.

Thermo-resistive devices and thermocouples are commonly used temperature sensors. As shown in Figure 4I, Temperature, thermistors are thermo-resistive devices whose resistance changes with temperature and can have a negative or positive temperature coefficient. Resistance temperature detectors (RTDs) are metal conductors with a resistance that increases with temperature.[30,31] Thermocouples are one of the most common sensors used in the industry and are formed by welding or crimping two different metal joints together. The voltage difference between the joints is used to measure temperature using the "Seebeck effect." However, the characteristic curve of a thermocouple is nonlinear in terms of voltage versus temperature because the Seebeck coefficient of a metal is not linear.[32] Because the temperature sensor is

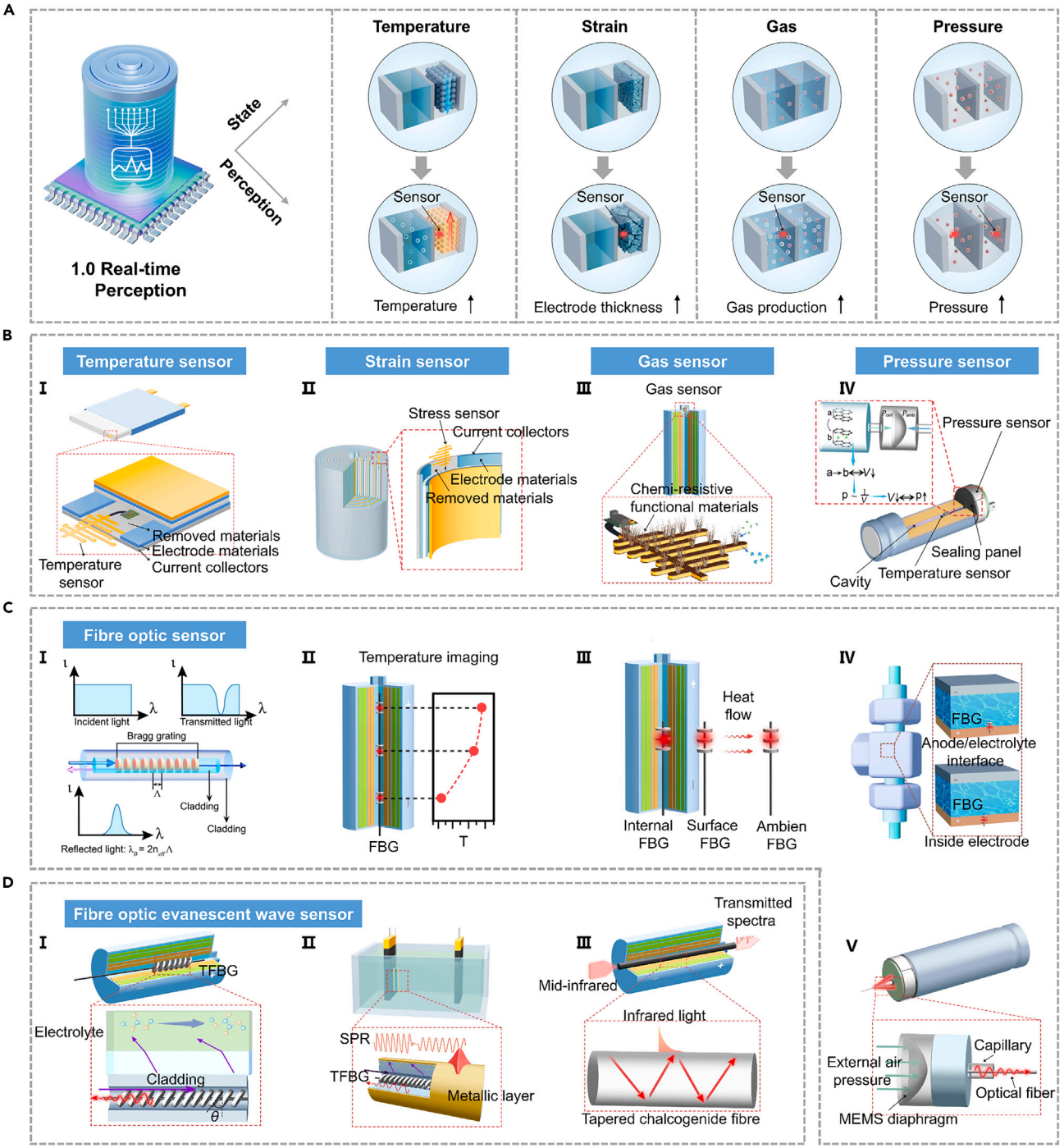


Figure 3. Perception principle application diagrams

(A) Smart batteries 1.0—real-time perception, various sensors sense the internal temperature, strain, gas, and pressure physical field information of batteries by internal implantation.

(B) Multi-point thin-film temperature sensor integrated into the electrode to determine changes in the internal temperature of the battery (BI); the strain sensor placed at the space provided by the cathode pole and located at the middle of the jelly roll radius to measure the real inner circumferential strain (BII); gas sensors based on chemi-resistive functional materials into the battery (BIII); temperature and pressure sensors are implanted inside the cylindrical battery to measure the internal gas pressure and internal temperature of the battery (BIV).

(C) Working principle of FBG sensor (CI), and the operation monitoring of internal temperature (CII), heat (CIII), strain (CIV), and pressure of batteries (CV).

(D) Fiber optic evanescent wave sensors are implanted inside the battery to *in situ* monitor RI (DI), ion activities (DII), and chemistry evolution (DIII).

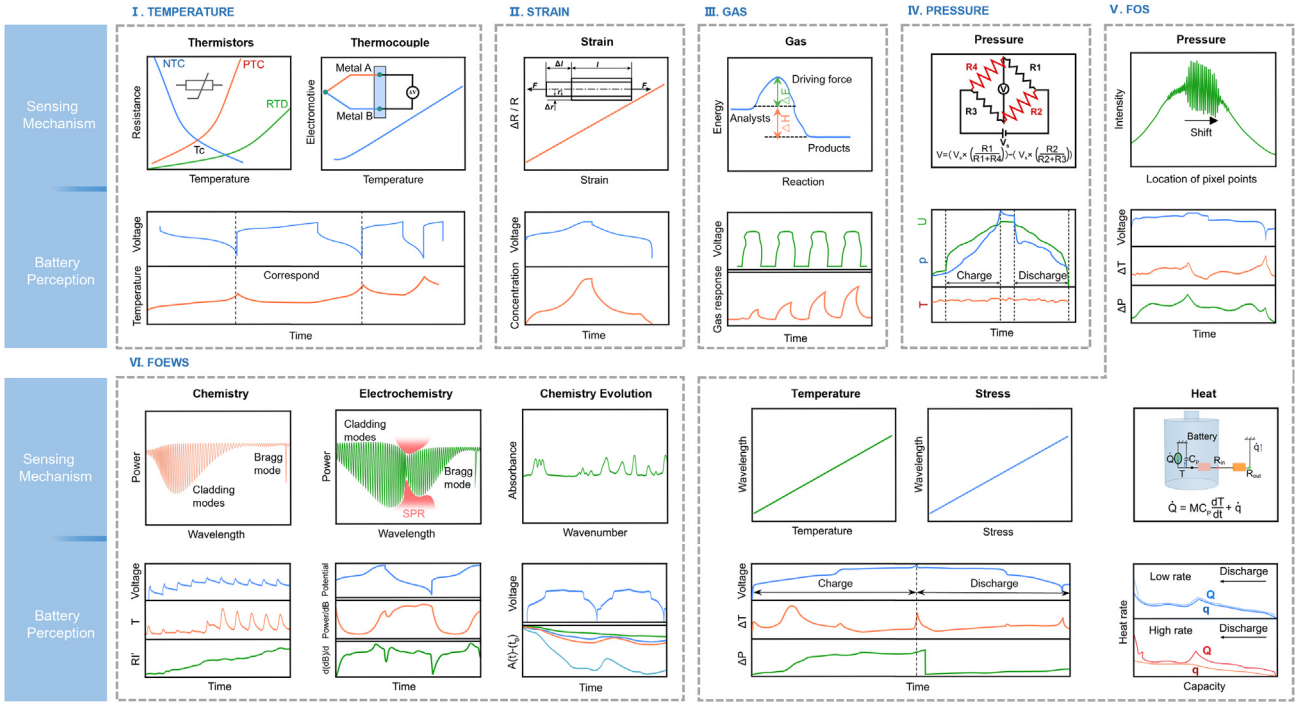


Figure 4. Sensing mechanism characteristic curves

Sensing characteristic curves of temperature (I), strain (II), gas (III), pressure (IV) sensors, FOS (V), FOEWS (VI), and the process of change with battery charge and discharge perception.

implanted inside the battery, the temperature feedback inside the battery changes in line with the charge/discharge curve, implying that the sensor responds successfully in sensing the internal temperature.

Strain sensor

The battery generates a great deal of stress during operation due to the volume expansion effect of electrodes. The resultant increase in internal stress squeezes the pole piece, leading to wrinkling or fracture of electrodes, which seriously affects the electrochemical performance.[33] Therefore, studying the mechanical failure of commercial batteries to obtain a more suitable method of measuring internal strain is needed. Implanting a strain sensor into the cylindrical battery is a successful method of monitoring the internal strain in the circumferential direction of the battery. Similar to the method of temperature sensor implantation, as shown in Figure 3BII, the strain gauge sensor is placed in the middle of the radius of the jelly roll and at the space provided by the anode[34] to achieve *in situ* measurement of the internal circumferential strain during the charging/discharging.

The resistive strain gauge is a relatively traditional resistance sensor used widely in the industry. It senses resistance-strain effect, i.e., the conductor or semiconductor material undergoes mechanical deformation under external forces, causing the resistance value to change with deformation (elongation or shortening).[35] As shown in Figure 4II, Strain, resistive strain has a linear relationship with the longitudinal strain over a wide range. Therefore, by applying the strain gauge to the test battery, the volume expansion/contraction of electrode as it is charged and discharged can be directly read from the strain gauge, assisting in the measurement of the evolution of the circumferential strain.

Gas sensor

Battery thermal runaway is a pre-production issue, where gas production is an important feature. As the temperature rises, the internal side reactions of battery produce more heat and gas, providing more accelerant for battery combustion and explosion to generate safety risks.[27,36] The current techniques that can monitor and analyze the gases inside the battery are mainly through some mass spectrometry-type techniques. These techniques include online electrochemical mass spectrometry (OEMS), differential electrochemical mass spectrometry (DEMS),[37] etc. to illustrate the correlation between voltage and gas concentration. However, these large devices require the design and use of specific electrochemical batteries, and the testing process is time delayed and cannot be applied to commercial batteries. The implantation of gas sensors inside the battery is obtained to be a solution of this problem. This allows for real-time monitoring of hazardous gas (combustible and toxic) output and concentration components. Appropriate warning thresholds can be set based on the thermal runaway characteristics of the battery, which can lead to early detection and warning. By placing the battery and various major gas-producing components of non-dispersive infrared (NDIR) gas sensors in a sealed tank,[38] the issue of measuring the internal gas of commercial batteries can be resolved. By measuring the intensity of the infrared light entering the infrared sensor, gas concentration inside the battery can be monitored and quantified without affecting battery operation, providing a basis for evaluating the battery side reactions. However, due to significantly lower gas concentrations measured externally compared with the internal environment of closed-system power devices, this approach is impractical in practical applications.

Embedding gas sensors based on chemi-resistive functional materials into the battery seems to be a feasible solution.[39] By utilizing techniques such as magnetron sputtering, gas-sensitive materials can be deposited onto the sensor surface and placed inside the pressure relief valve. When the sensor absorbs gases generated internally within the battery, it causes a change in resistance (Figure 3BIII).

Gas sensors based on chemi-resistors typically involve three steps: the surface gas molecule capture process, the surface reaction process, and the charge transfer process. These sensors mostly use sensing elements based on single materials or transduction mechanisms, with their inherent sensing activity typically acting as a driving force to stimulate the sensing effect of the target gas (Figure 4III, Gas). There are typically five common sensing modes for the gas detection layer.[40–44] Among them are the following: (1) catalytic effect: gas analytes achieve high responsiveness, rapid rates, and low operating temperatures through chemical/electronic sensitization between the gas analyte and the modified catalyst. (2) Charge transfer effects: this encompasses swift charge transfer processes between decorations and host materials to modify the conductivity of hybrid materials. (3) Regulation of charge transport: manipulating the sensing properties by changing carrier concentrations, transportation mode, and/or pathways of charge transport. (4) Semiconductor heterojunctions: building semiconductor heterojunctions (n-n, p-n, p-p, and p-n-p), connected by energy band structure due to the alignment of their fermi level, while excluding catalysis, charge transfer, and charge transport. (5) Surface probes: utilizing selective gas physical adsorption based on surface probes[39] and the examples of gas sensing layer materials and specified gases shown in Table S1. Currently, most gas sensors are based on single materials or sensing mechanisms and may not achieve ideal sensor performance. To enhance sensing performance, it is possible to modify the sensor behavior by designing complex compositions and novel structures for the sensing layer, accurately targeting multiple physical/chemical processes.

Pressure sensor

As mentioned above, gas by-products are an inevitable issue that arises due to side reactions during battery degradation. The accumulation of these gases increases the internal pressure of the battery, thereby damaging the contact between the particles of electrode materials, leading to an increase in resistance, battery deformation, accelerated aging, and major safety hazards.[45] It is important to monitor the internal pressure of the battery in real-time to understand battery aging behavior. There are some cases for measuring the internal battery pressure and gas composition, such as online mass spectrometry and multichannel *in situ* barometric device coupled with DEMS.[46,47] These methods can only be used for research purposes due to the size of device, making it difficult to implement in practice. Therefore, the development of a miniaturized, implantable battery internal pressure sensor is necessary.

It is feasible to implant temperature and pressure sensors within commercial cylindrical LIBs to measure the internal pressure and temperature during the cycles.[48] The ideal gas approximation is typically allowed when the gas pressure is less than the critical pressure based on the ideal gas law. It may be believed that the critical pressure of the gas inside the battery is unaffected by temperature because it is much greater than the pressure inside the battery. Assuming that the total volume inside the battery is constant, the electrode expansion decreases the gas volume and increases the gas pressure.[49] The schematic diagram of the measurement device represented in Figure 3BIV demonstrates that an increase in the amount of Li^+ embedded during charging and discharging causes an expansion in the electrodes' volume, ultimately leading to a reduction in the battery's intrinsic volume. By employing a Whiston bridge as a sensor circuit, the pressure sensor can detect the increase in internal pressure as a result of the voltage signal generated by the piezoresistive (deformation) effect of the resistor.[50] The characteristic curve outcomes are evident, with identical trends observed for pressure and voltage curves during charging, followed by a decrease in pressure reaching a constant value during a pause, and eventually a further decrease while discharging (Figure 4IV, Pressure).

Fiber optic sensor

With the rapid development of optical fiber communication technology, optical fiber sensor technology has also grown rapidly. It is a sensing technology that uses optical waves as carriers and optical fibers as the medium to sense and transmit external measured signals.[51] Fiber Bragg grating (FBG) optical sensor is a relatively novel and evolving area of research for implanted sensors due to its small size, light weight, high corrosion and interference resistance, high sensitivity, and high resolution.[17,52] It senses and transmits the measured signal by correlating the wavelength dependence of the signal (λ_B) with the local temperature (T) and the stress (ε) (Figure 3CI).[53] When the sensor is subjected to external influences, changes in neff and Λ cause a shift in the reflected peak, which is termed as the Bragg wavelength (λB), defined as

$$\lambda_B = 2un_{eff}\Lambda. \quad \text{(Equation 1)}$$

The FBG sensor exhibits that the change in temperature (ΔT) and stress ($\Delta\varepsilon$) is linearly proportional to the Bragg wavelength shift ($\Delta\lambda_B$) when the sensor is subjected to battery influences as illustrated in Figure 4V, FOS: Temperature and Stress. Under practical operating conditions, FBG can not only detect the temperature gradients of electrochemically driven in batteries but also detects the effect of overcharge or external short circuit on temperature (Figure 3CII).[54] Integrating FBG optical sensors into commercial 18650 batteries allow for the collection of rich real-time chemical and thermodynamic information during operation to receive a fully parameterized thermal model.[55] A simplified zero-dimensional thermal equivalent circuit

(Figure 4V, FOS: Heat) and a heat balance equation are used to convert the collected data into heat as the temperature of the FBG changes during the battery cycle in the external environment, on the battery's surface, and inside the battery by placing three FBG sensors (Figure 3CIII),

$$\dot{Q} = MC_p\frac{dT}{dt} + \dot{q} \quad \text{(Equation 2)}$$

where M, C_p, and T denote the mass, specific heat capacity, and average battery space temperature of the battery, respectively, and $\dot{Q}$ includes all the heat generation processes, and $\dot{q}$denotes the heat dissipation to the environment.[56] It is noteworthy that when the battery is discharged at a high rate, the heat generated is substantially greater than the heat dissipated to the environment, indicating that accumulated heat is a key contributor to battery heat production.

Moreover, the feasibility of using FBG sensors for monitoring the electrode/electrolyte interface stress in all-solid-state batteries was confirmed.[57] When the deformation of the electrode applies stress to the FBG, the change of the optical signal occurs because of strain. The FBG implanted into the middle of the electrode exhibits a peak shift and splitting of the spectrum, which is mainly attributed to the accumulation of transverse and axial stresses. The FBG was implanted into the battery at the middle of the electrode/electrolyte and the upper surface of the electrode, respectively, and both exhibited similar trends in the change of stress $\Delta\varepsilon$, as depicted in Figure 3CIV.

Fiber optic sensors also offer a precise method for measuring the internal gas pressure of batteries. Utilizing compact fiber optic sensors, a micro-electro-mechanical system (MEMS) creates a Fabry-Perot (F-P) interferometer at the tip of the optical fiber, enabling high-sensitivity pressure measurements while mitigating temperature interference.[58] This technology enables the absolute monitoring of gas pressure within commercial LIBs, providing insights into gas production and pressure fluctuations (Figure 3CV). According to the F-P interference principle, an interference signal representing the F-P cavity length "h" is generated when the reflected beams from the two end faces of the vacuum cavity intersect within the optical fiber. As the pressure on the sensor gradually increases, the cosine Gaussian signal gradually shifts toward higher pixel positions (Figure 4V, FOS: Pressure). The cavity length exhibits a robust linear correlation with pressure, and the sensitivity to temperature variations is minimized.

Fiber optic evanescent wave sensor

As mentioned above, the use of FBG sensors enables the monitoring of various parameters such as temperature (T), strain (ε), and pressure (P) in real-world conditions, offering high sensitivity and spatial resolution. However, in such a configuration, the optical fiber, as a genuine waveguide, is completely transparent to the surrounding environment. To acquire rich chemical information related to the electrode materials or electrolyte properties inside the battery, it is necessary to overcome the limitation of light being confined within the fiber core. This limitation can be circumvented by modifying the fiber's structure. By incorporating a hollow-core fiber probe into laboratory-scale pouch cells and integrating it with Raman spectroscopy, it becomes feasible to trace the changes in the electrolyte during cycling.[59] Nevertheless, this approach relies on external systems (e.g., pumps and tubing), which impede the continuous monitoring and tracking of alterations in the electrode materials themselves.

By modifying the design of the optical fiber, it is possible for light propagating along the fiber's axis to partially escape from the core. This optical sensing principle relies

on the interaction of the evanescent fields at the boundary between the optical fiber and the surrounding medium, as seen in fiber optic evanescent wave sensors (FOEWS). Tilting the grating plane of FBGs can generate a distinct type of sensor known as a tilted FBG (TFBG). In contrast to a standard FBG, the grating plane of a TFBG is rotated at a specific angle around an axis perpendicular to the optical fiber axis, ensuring coupling between a single forward-propagating core mode and multiple backward-propagating cladding modes through a phase-matching relationship.[60] When the refractive index (RI) or dielectric constant of the external medium experiences any disruption, light from the forward-propagating core mode couples with the backward-propagating cladding modes (indicated by purple arrows), leading to multiple resonances (tilt angles) in the collected spectrum (Figure 3DI). These discrete cladding mode resonances exhibit shifts in wavelength and/or amplitude in response to disturbances, as the evanescent fields of the cladding modes can penetrate the surrounding medium, such as an electrolyte, and interact with it (Figure 4VI, FOEWS: Chemistry).

Furthermore, significant advancements have been made in the integration of electrochemical surface plasmon resonance (SPR) fiber optic sensors into batteries, enabling *in situ* detection of chemical and electrochemical events without disrupting normal battery operation.[61] In this context, the plasmonic sensor can be envisaged as an optical fiber coated with a thin layer of metal that can induce SPR. This resonance is inherently sensitive to changes in the RI of the surrounding environment. Given that alterations in RI are associated with physical and chemical changes, SPR sensors can be employed to monitor such changes within a range of several hundred nanometers to 1 μm around the sensor. One approach involves imprinting a TFBG into commercial single-mode optical fibers and coating it with a nanoscale gold film to efficiently excite SPR, as illustrated in Figure 3DII.[62] The phase velocity and attenuation distance of SPR waves at the optical fiber sensor's surface near the battery electrode are influenced by minor local variations in ion transport. This leads to changes in the position and shape of the measured TFBG-SPR spectral features in the reflection (Figure 4VI, FOEWS: Electrochemistry). Compared with using a bare TFBG, this method provides higher RI sensitivity.

Although the acquisition of RI data offers potential advantages, it is important to acknowledge the presence of uncertainties stemming from specific electrolyte degradation or variations in salt or solvent concentrations. One potential approach to address this limitation draws inspiration from the field of spectroscopy. The utilization of operando infrared fiber evanescent wave spectroscopy to monitor electrolyte evolution within batteries under real-world operating conditions has been substantiated.[63] This method involves the use of sulfur-based (sulfides, selenides, and tellurides) glass fibers that traverse the central void of a 18650 sodium-ion battery (Figure 3DIII). These fibers are connected to an infrared optical signal generator and detector, enabling electrochemical cycling of the battery with varying electrolyte compositions. Through the analysis of spectra collected during the formation and extended cycles, this approach facilitates the identification of chemical species and unveils the decomposition mechanisms of the electrolyte and additives during cycling (Figure 4VI, FOEWS: Chemistry Evolution). As a result, it offers invaluable insights into the growth and characteristics of the solid-electrolyte interface, solvent dynamics, and their intricate interdependencies.

Interactive feedback smart batteries

According to its unique function categories, dynamic responsive smart batteries can be categorized into self-protection, self-healing, self-charging, self-adaption, and self-switching (Figure 5A). The application of smart materials enables the battery

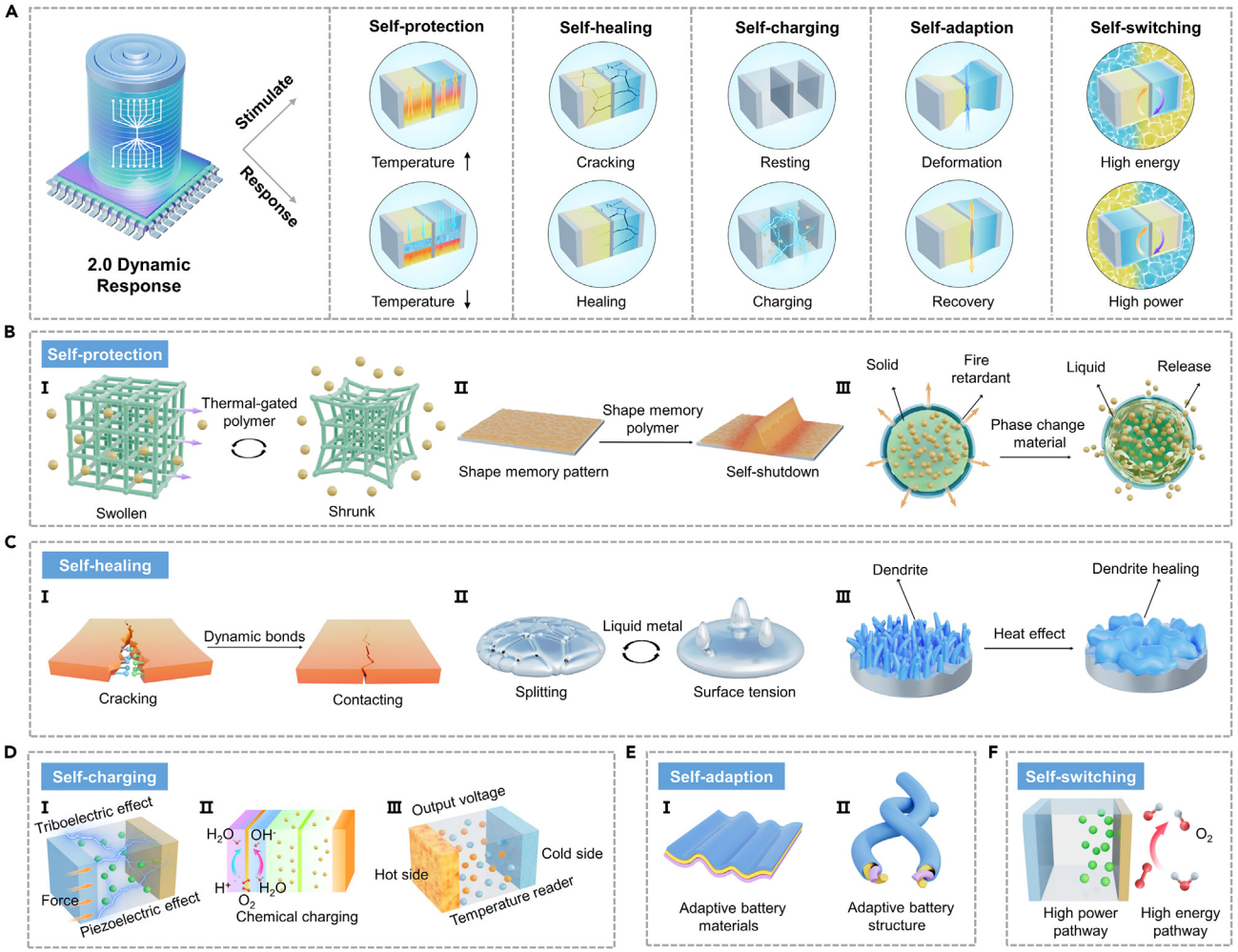


Figure 5. Feedback principle application diagrams

(A) Smart batteries 2.0—dynamic response, self-protection, self-healing, self-charging, self-adaptive, and self-switching functions in response to internal and external environmental stimuli.

(B) Self-protection strategies: thermal-gated polymer electrolyte reversibly opens/closes ion channels in response to temperature changes (BI); above the response temperature the shape-memory-polymer collector reverts to its original shape causing the circuit to break (BII); the phase-change material encapsulated in the microcapsules melts and releases the flame retardant when the temperature reaches the melting point (BIII).

(C) Polymers with dynamic bonding dominance (CI), liquid metals with high surface tension (CII), and surface migration of lithium atoms enhanced by thermal effect (CIII) realize self-healing.

(D) Collect the mechanical energy (DI), chemical energy (DII), and heat energy (DIII) in the surrounding environment and convert into electric energy to realize self-charging.

(E) Strategies for the implementation of self-adaption functions include the design of flexible cell materials (EI) and flexible/stretchable battery structures (EII).

(F) Electrode materials with multi-path electron storage and transport mechanisms enable self-switching seawater batteries.

to interact with the environment and provide various special functions, allowing it to respond and provide feedback in a timely manner to both internal and external stimulus and changes. As a result, this type of battery is able to function effectively in special environmental conditions.

Self-protection

Many organisms possess unique self-defense skills in nature, such as the gecko's ability to cut off its tail. In a broader sense, self-protection can be defined as the

instinctual or conscious actions taken by a system to protect itself from harm. It encompasses a range of behaviors, mechanisms, or strategies aimed at ensuring the continued existence, function, or integrity of the relevant entity. In the context of batteries, battery self-protection refers to a set of smart materials and measures meticulously designed to ensure its performance, safety, and lifespan under specific conditions. Simultaneously, it aims to prevent significant damage or hazards that may arise from abnormal situations or improper usage. More precisely, self-responsive sensitive materials employed in batteries have the capability to autonomously make decisions through feedback mechanisms. These decisions result in conformational and chemical alterations that ultimately ensure battery safety.[64] High thermal sensitivity and controllable thermal response temperature make thermosensitive materials ideal "smart gatekeepers" to safeguard against battery safety issues. Once the temperature signal exceeds a particular limit, a further thermal response can permanently or reversibly cut off, rendering the battery dormant and preventing safety accidents. These thermal materials can be used directly as battery materials or as auxiliary materials inside the battery.

Thermosensitive materials are often used in the synthesis of electrolytes, when applied directly as battery materials. Under thermally stimulated circumstances, thermally responsive electrolytes close the ion channels between cathode and anode, isolating electrodes from the electrolyte. When the temperature inside the battery hits a dangerous level (70°C), ion transport is irreversibly halted, thereby shutting down the battery and eliminating the possibility of combustion and explosion.[65] There are also thermally responsive electrolytes with the ability to reversibly open/close ion channels, such as a thermal-gated poly(N-isopropylacrylamide) (PNIPAM)-based hydrogel coated on a glass fiber separator.[66] This PNIPAM-based separator shows the evolution of the pore structure from open to closed state as the temperature increases and also shows a shift in surface wettability from hydrophilic to hydrophobic, which can inhibit the migration of electrolyte ions between the separator. The ion gate battery has similar cycling performance to a normal battery in the "open" state, and when the ion gate is in the "closed" state, it can effectively shut down the battery and achieve self-protection (Figure 5BI). Additionally, shape-memory polymers (SMPs) are smart that recover from pre-programmed temporary shapes to initial shapes after receiving a thermal drive beyond response temperature.[67] A new smart shape-memory current collector (SMCC) can prevent the battery from thermal runaway while the inside is overheating. SMCC is made by depositing copper on a thin film of micro-patterned shape memory, showing ideal conductivity under room temperature and becoming insulating at superheat temperature. When the surface's collapsed micro-pattern reaches the phase-change temperature, it will recover its initial shape, blocking the connection between the dense metal layer and the circuit disconnector and rapidly achieving self-shutdown before the battery explodes (Figure 5BII). Microencapsulation is a strategy that can be used as an accessory material. For instance, Hu et al. encapsulated phase-change materials (PCMs) and triethyl phosphate (TEP) flame retardants in microcapsules to prepare thermally responsive novel separators (Figure 5BIII).[68] The microcapsules prevented the flame retardant from dissolving directly into the electrolyte, minimizing the adverse effects on battery performance and resolving electrochemical incompatibilities between the TEP and electrodes. In case of internal temperature rising to the melting point, the PCM melts and releases the TEP in time, thus improving battery fire safety.

Thermally sensitive materials have been advanced mainly for self-protection against thermal runaway. However, the same concept can be applied to other mechanisms such as electrochemical property changes. Figure 6I, Self-protection, illustrates the

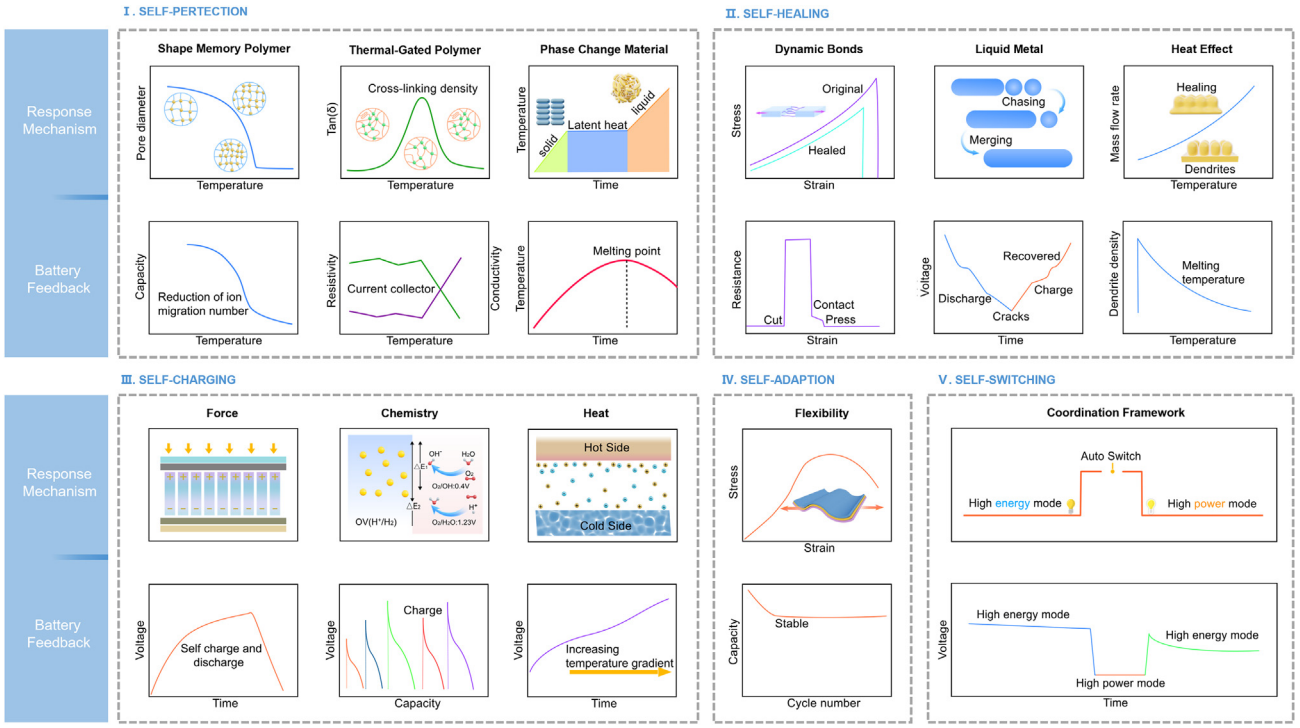


Figure 6. Response mechanism characteristic curves

Response characteristic curves of self-protection (I), self-healing (II), self-charging (III), self-adaptive (IV), and self-switching (V) smart functions and feedback changes of battery electrochemical performance.

following response mechanisms: (1) thermal-gated polymers with porous structures can be controlled by temperature. At room temperature, the thermal-gated polymer-based separator pores absorb solution electrolyte while in a highly expanded state. As temperature increases, the electrolytes are gradually released from the hollow structure, causing the capacity of the battery to reduce until the pore closes and the battery shuts down. (2) SMP can regulate the glass transition temperature positively by increasing cross-link density. At high temperatures, the SMP pierces the covered copper layer, resulting in a cracked trench where electrons are unable to migrate freely across the shape-memory-polymer-based collector, causing circuit disconnection. Simultaneously, the collector fluid's conductivity and resistivity change drastically, indicating that it is in a non-conductive state. (3) The PCM undergoes a solid-liquid phase transition under thermal stimulation, which increases ion mobility. As the internal temperature of the battery reaches the melting point of the PCM, the flame retardant is promptly released from its encapsulation. This significantly mitigates the adverse effects on battery performance, while the heat absorption reaction of the phase-change process leads to a decrease in internal battery temperature.

Self-healing

In response to damage to organ functions, living organisms are born with a set of repair mechanisms. Self-healing is a common natural phenomenon in biological systems. Similarly, physical/chemical fatigue and damage problems of the intrinsic material during aging and after mechanical shock are inevitable in batteries. Thus, the injection of self-healing capability into the battery body by simulating organic organisms is considered an effective way to improve this situation.[69] Battery self-healing

refers to a specific battery design and technology that enables the battery to take automatic measures to repair or restore its performance when it is damaged, aged, or experiences other abnormal conditions. This typically involves the use of special self-healing materials or techniques to repair physical or chemical damage within the battery, such as microcracks in electrode materials or degradation of the electrolyte. In doing so, it assures the battery's functionality and extends its longevity. The response mechanism of self-healing function in batteries has various types, and the self-healing repair achieved by the strategy is usually classified as polymer based on dynamic bonding dominance, fluid-liquid metals, and temperature thermal effects.[70]

The lifetime of batteries is limited by the mechanical fracture dilemma during cycling. Electrochemical reactions in batteries can cause structural changes such as electrode cracking and pulverization, resulting in degradation and damage. In order to combat this issue, self-healing polymers (SHPs) that utilize dynamic covalent and non-covalent bonds are employed. SHPs display various types of dynamic bonding mechanisms, including Schiff base reactions, borate ester bonds, oxime bonds, hydrogen bonds, ionic interactions, and metal-ligand coordination,[71] to reversibly restore the molecules to original state after impairment (Figure 5CI). With this, specific healing agents are added during the manufacturing process or applied directly to the battery materials to create self-healing smart batteries.

The use of SHP binders is the main application strategy to obtain electrode stability. Significant mechanical degradation, including significant fracture and particle shedding, has been observed during lithium alloying of the silicon anode. Cycle life can be improved by coating the particles with a thin layer of hydrogen-bonded oriented SHP.[72] The SHP binder has good mechanical stretching properties, resulting in contact between the shrinking silicon particles and polymer binders that can accommodate the potential expansion of the electrode while avoiding large non-healable fractures. When unavoidable small fractures occur in the SHP binder, the effect of hydrogen bonding re-formation enables the fracture surface to shrink continuously resulting in tighter contact and complete healing.

Introducing the self-healing ability of electrolytes can be a good solution to combine mechanical strength and ionic conductivity, also solve the problems of solid electrolyte interface (SEI) fracture/dissolution and dendritic growth.[73,74] It is also important to design high-performance solid-state batteries with self-healing interfaces using SHP,[75] which can further prevent the occurrence of interfacial side reactions, relieve the stress generated by dendritic growth and automatically repair the cracks generated at the interface due to the expansion of the electrode volume.

In this scenario, the toughness of SHPs is the main characteristic parameter because it determines the material's ability to recover its original and healed toughness.[71,76,77] Therefore, by assessing the recovery of toughness in SHPs applied to batteries, it is possible to evaluate the effectiveness of this technology in enhancing battery lifespan. When applied to batteries, the most obvious characteristic is the change in resistance, which becomes infinitely high when the battery is broken. On the other hand, the resistance decreases when it is reconnected and healed (Figure 6II, Self-healing: Dynamic Bonds). However, for materials that require external conditions such as light and heating to activate their self-healing capabilities, additional components such as heating elements or controllable LED lights need to be provided to deliver the necessary stimuli for self-repair. Another feasible approach is the integration of sensors and control systems. When a damage event is detected,

the battery can communicate with external devices or networks to trigger the required conditions for self-healing, thereby enhancing the adaptability and autonomy of the battery system's self-repair.

Additionally, the use of liquid metal (LM) electrodes has become a prevalent application in batteries, specifically for molten liquid alloys, including alkali metals such as Na-K and Na-Cs alloys, and post-transition metals such as Ga-In and Ga-Sn alloys.[78,79] LM is an effective self-healing strategy for electrode mechanical degradation because of its liquid flow deformability and excellent surface tension, making it a better candidate for self-healing (Figure 5CII). Electronegative metals, such as Li, K, Na, Ca, and Mg, can be used as anode materials, and electropositive metals, semimetals, or elements within Group VIA on the periodic table, such as Bi, Te, Ga, Pb, Sn, Sb, and S, as cathode materials, which have been shown to be applied as LM electrodes.[80]

The self-healing response mechanism is schematically shown in Figure 6II, Self-healing: Liquid Metal,[81] where the LM stream splits into many micro-droplets during charging and discharging. The larger micro-droplets can swallow the smaller ones due to the strong surface tension and eventually form a larger droplet due to collision and agglomeration. Therefore, based on this operational behavior, the application of LM electrodes can exhibit excellent cycling performance by over-modifying the surface energy and morphology to achieve dendrite-free growth. Although LMs have practical applications, they require careful handling due to the fluidity and reactivity. LMs are chosen based on their specific properties, and they can be highly advantageous in situations where traditional rigid conductors or materials are not suitable.

Thermal effects generated by external current stimulation can repair and regenerate internal dendritic growth, in addition to the introduction of self-healing materials. It is widely believed that the dendrite growth will gradually increase with the increase of current density.[82] However, it was found that the high current density induced self-heating in Li dendrites through the Joule effect, when the plating/stripping current density was increased to ~9 mA cm^{-2}, which enhanced the surface migration of Li.[83] This surface diffusion repaired the dendrites achieving a smooth Li metal surface, as well as a high Coulombic efficiency (Figure 5CIII). Figure 6II, Self-healing: Heat Effect, demonstrates the tendency of the migrating mass flow rate of Li atoms with temperature. As the Li melting temperature is reached, the migrating mass flow rate sharply increases, whereas the dendrite density decreases accordingly.

Self-charging

In daily life, there are many different sources of energy, such as mechanical, chemical, thermal, optical, biological, and electrical energy, which can be transformed into each other through physical effects or chemical reactions. To make use of these plentiful, intermittent, and randomly distributed energy sources, environmental energy can be converted into electrochemical energy and used as a self-powered sustainable solution for electronic devices. This can be accomplished by harvesting energy from the surrounding environment and outputting the power to electronic devices. Battery self-charging refers to the ability of a battery system to acquire energy from its surroundings or other sources and use this energy for self-recharging or sustaining the battery's charge state, without the requirement for external charging equipment. This capacity often involves the utilization of renewable energy sources such as solar power or mechanical motion, or the utilization of internal chemical reactions or thermal energy to enable the battery's autonomous recharging (Figure 5D).[84–88] The self-charging function can extend the battery's lifespan, offer convenience, and reduce reliance on external power sources, making it particularly

significant in various applications such as remote sensors, wearable devices, and certain portable electronic devices.[89]

The mechanism of mechanical energy in energy harvesting processes is mainly attributed to two phenomena: the triboelectric effect and the piezoelectric effect.[84] The triboelectric effect is characterized by contact electrification and electrostatic induction when two materials rub against each other. This causes the outermost electrons of the materials to receive enough energy to cause charge transfer, which maintains electrostatic equilibrium by electron flow through an external circuit.[90] Piezoelectric effect, on the other hand, occurs when an external force is applied to a piezoelectric material, causing its deformation in a particular direction and the occurrence of polarization inside the material. This leads to the development of opposite positive and negative charges on its two opposite surfaces (Figure 6III, Self-charging: Force).[84] There is also the chemical energy self-charging mechanism, which collects energy from the surrounding environment to induce redox reactions or ion migration at the electrode, converting chemical energy into electrical energy (Figure 6III, Self-charging: Chemistry). Finally, thermal charging induces diffusion of holes and electrons/ions from the hot side in a conductor or semiconductor connected between two electrodes, resulting in the creation of a thermal potential and a direct current in an external circuit (Figure 6III, Self-charging: Heat).

Self-adaption

The rapid development of flexible wearable electronics has necessitated the creation of adaptive smart batteries that can meet the device's biocompatibility and flexibility requirements.[91] Self-adaption refers to the ability of a battery's design and construction to possess flexibility and adaptability, allowing it to accommodate various shapes, sizes, curves, or surfaces in different application environments. Traditional rigid batteries are not suitable for this purpose, flexibility in battery materials or the flexible/stretchable design of battery structures are key features (Figure 5E).[92,93] These features enable the battery to return to its initial design shape through its physical and chemical properties after deformation. The ability to regulate the volume changes caused by materials during the entire electrochemical process and withstand the stress changes caused by mechanical deformation of the device are both benefits of adaptive characteristics. Based on this adaptability feature, it helps the battery seamlessly integrate into its application environment, providing greater design flexibility, and enhancing the comfort and performance of electronic devices.

These strategies allow normal battery operation under complex conditions. Adaptive flexible batteries rely on flexible electrodes such as flexible collectors (carbon-based materials, such as carbon cloth, graphene paper, carbon nanotube paper, etc.) and flexible active materials, which include polymers, LMs, and array structures.[94,95] Additionally, the design of flexible batteries at the battery structure level is critical for assembling different strategies into a flexible battery, thus adapting to the deformation by changing the shape at the macroscopic level. These strategies can be divided into one-dimensional columnar structures, two-dimensional (2D) planar structures, and three-dimensional (3D) structures, which mainly include four major strategies for designing flexible batteries, namely, wave structured batteries, foldable batteries, fibrous batteries, and intrinsically stretchable batteries.[92,96] Importantly, mechanical stress-strain affects the kinetics of electrochemical processes such as mass transfer, charge transfer, interfacial reactions, and phase changes,[93] thus affecting the performance, capacity, and efficiency of the battery and the ability to maintain good electrochemical stability within a reasonable strain range is the primary criterion for evaluating batteries adaptivity (Figure 6IV,

Self-adaption). A battery device that shows significant mechanical strength and plasticity without dramatic decay of electrochemical properties under applied strain/stress has good self-adaptability.

Self-switching

In order to realize the combination of high energy and high power, which is required under complex working conditions, batteries need to possess another precise control capability-automatic switching of power supply modes. This automatic selection of the appropriate output mode is made possible by the smart battery's ability to adapt to different external environments. The self-switching capability of batteries refers to the battery system's automatic adjustment of its output power and energy density to meet varying application requirements. High-power batteries can deliver higher currents for situations requiring instantaneous high energy output, whereas high-energy-density batteries possess greater operation life, providing stable energy output for longer durations. This self-switching feature allows the battery to automatically switch between high-power and high-energy-density modes based on the current usage demands, maximizing the acceleration ability and endurance. Marine organisms utilize respiration as a strategy to combine long endurance and high-power modes.[97] Likewise, a dual-mode seawater energy extraction system with automatic switching function can be achieved by using cathodes with multivalent coordination skeletons (Figure 5F).[98] This system employs multi-path electron storage and transport mechanisms, allowing it to automatically select different solutes in seawater as electron acceptors to provide higher energy densities and power densities, respectively (Figure 6V, Self-switching). Importantly, it can potentially inspire additional power supply modes for future switches.

Self-decision-making smart batteries

The fourth industrial revolution, powered by digital technologies, such as the Internet of Things (IoT), big data, and AI, is vigorously sweeping the world.[9] With the arrival of the era of intelligence, smart decision-making abilities are highly pursued to provide the mainstream trend of electrification and intelligence. As a result, a challenging issue of increasing trade-offs between reliability, efficiency, and safety has thus been faced by the battery sector. Although traditional BMS have been employed to monitor and protect batteries from faults, which suffer from unreliability in real-time fault diagnosis, mainly due to algorithm model inaccuracies, limited computing power, and data storage insufficiencies.[99,100] To ensure reliable monitoring of battery operation status, real-time estimation of current battery health information (e.g., aging trajectory and remaining useful life) is critical. Such information facilitates the adoption of various reliable battery control strategies to ensure battery safety and prolong battery life. In order to better monitor battery status, visualize data in real time, enable more accurate and reliable battery prediction and diagnosis, and ultimately enable more effective autonomous control and system decision optimization, big-data-driven self-decision-making smart batteries utilize these technologies. Based on this, a self-decision-making framework for the entire life cycle of smart batteries is provided in Figure 7.

In order to solve the challenges of bottlenecks in battery materials optimization and various problems caused by the complex structural changes of batteries, such as non-uniform internal temperature and pressure, and inconsistencies in the composition of a single battery that are difficult to manage effectively, digitally driven battery design is gaining widespread attention and application in the industry nowadays.[12,101] Modeling and measurement are the basis of battery digitization, which is essential to determine the current state and predict the future state of batteries with high accuracy.

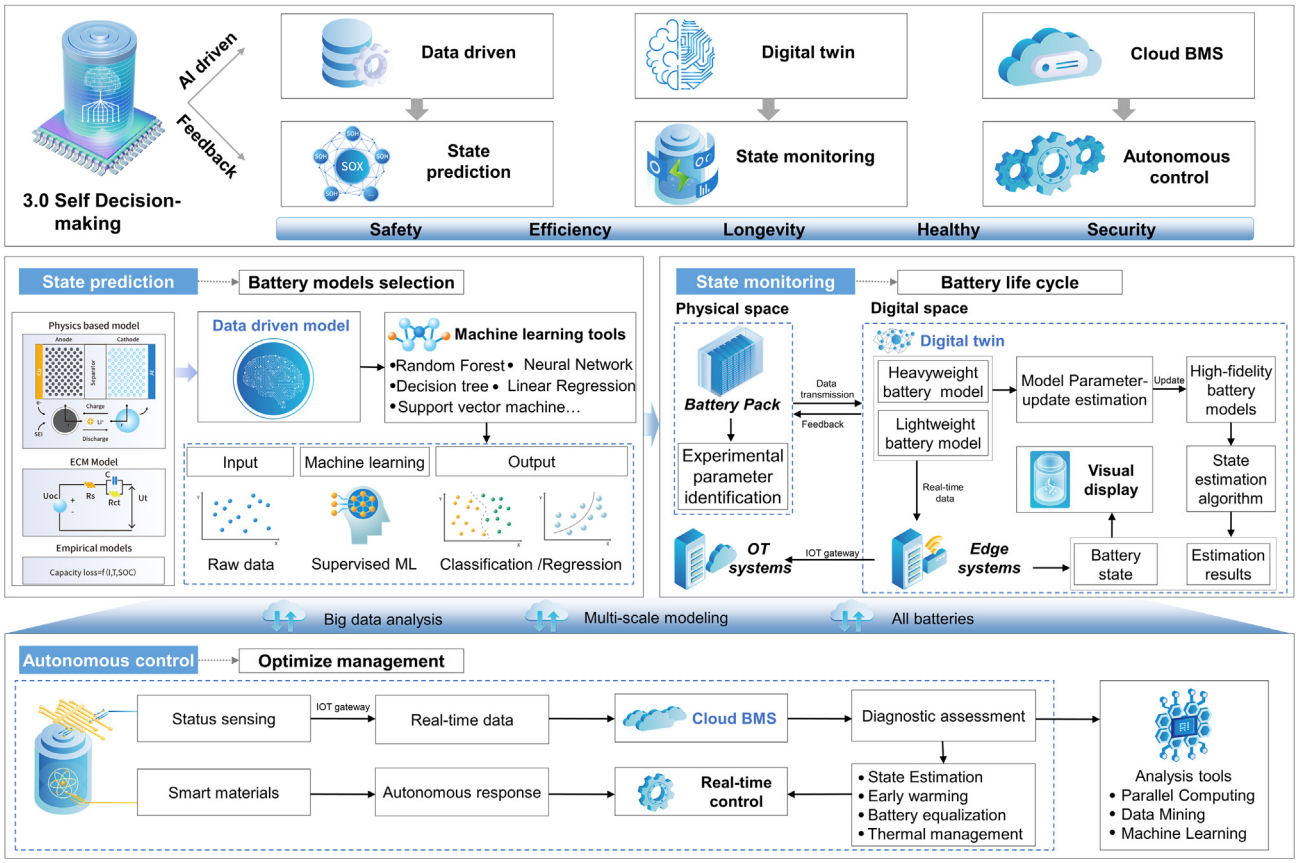


Figure 7. Self-decision-making framework

Smart batteries 3.0—self-decision-making, utilizing big data, digital twin, and cloud BMS technologies to achieve autonomous decision-making for smart batteries. Simulation data from multi-scale modeling and experimental data from high throughput are fed into machine learning algorithms to predict SOX; virtual digital twin models mapped to physical entities are constructed based on physical and data-driven models to accurately predict and diagnose battery status; cloud BMS comprising IoT components and cloud infrastructure to collect large amounts of multi-dimensional spatiotemporal data, ultimately enabling more effective autonomous decision control.

The existing methods used to estimate the battery state SOX, including SOC, state of energy (SOE), state of power (SOP), state of function (SOF), SOH, and remaining useful time (RUT), utilize physical models, equivalent circuit models, and empirical models.[102,103] However, the main challenge of existing battery models is that it is difficult to strike a proper balance between model fidelity and computational complexity. Data-driven models based on ML techniques have great potential to handle complex nonlinear functions, making them advantageous compared with traditional mathematical models. ML methods also include many kinds of methods such as Random Forest, Neural Network, Decision Tree, Linear Regression, and Support Vector Machine.[104,105] Choosing the most appropriate ML method depends primarily on the type of problem, followed by the size, quality, and nature of available data and the physical interpretability of the desired model. Simulation data from multi-scale modeling and experimental data from high throughput are fed into the ML algorithm to predict SOX. Importantly, the predicted values can be served as a guide for refining the accuracy of subsequent simulations or experiments.

DT technology has been proposed due to continuous technological breakthroughs in various fields such as models, algorithms, IoT, and big data.[106] Based on the above-mentioned physical model and data-driven model, a DT model can be built,

i.e., a virtual digital model mapped to a physical entity. This model can be used to map and simulate the entire life cycle of a physical battery in multiple dimensions (multi-physics, multi-scale, and multi-probability), especially useful in battery state estimation, fault diagnosis, and safety warning.[107] The physical and digital spaces are interactively coupled and co-evolved, which is ideal for application to battery systems with complex nonlinear characteristics.[108–110] The implementation of the DT architecture for batteries in an IoT network architecture is presented to define a process of systematically moving from a battery model to a battery DT function. The first parameter identification of the battery requires datasets from sensor measurements and the experimental environment of the battery, which is the basis for the battery DT parameterization. When it comes to the actual implementation of the framework, the battery model becomes an important tool for the functional drive. Importantly, to achieve a high-fidelity battery model, it is crucial to update the model parameters periodically during operation due to the different stages of the battery life cycle, causing changes in the operating state. Thereby, accurate and rapid prediction of the battery's full life cycle state can be achieved based on algorithmic processing of the data. Furthermore, it also facilitates communication between physical entities and digital virtual models of the battery for smarter control along with the ability to use IoT to run DT models in the cloud.

The BMS monitors the state parameters of the battery and performs necessary analysis and calculations for monitoring the battery state and managing safety aspects. In order to achieve more advanced intelligent autonomous decision-making, smart batteries implanted with multiple types of sensors, i.e., real-time sensing smart batteries, offer more possibilities for BMS to perform more accurate condition estimation and active control. However, driven by the cloud computing IoT, there is a higher requirement for the quantity and quality of data. The concept of cloud-based intelligent BMS design is proposed to address the limitations of traditional BMS data storage and computing power.[111–113] Cloud BMS consists of IoT components and cloud infrastructure, providing unlimited storage capacity and strong computing power.[112,114] It enables real-time monitoring and visualization of battery performance by collecting large amounts of multi-dimensional spatiotemporal data. For IoT components, a high-speed and reliable internet connection, such as 5G, plays a pivotal role in facilitating real-time data transmission between battery systems and the cloud. Furthermore, a Raspberry Pi can function as a compact single-board processing unit for data exchange and communication. Battery measurement data can be transmitted from the "edge" unit to IoT components through the controller area network (CAN) protocol. Subsequently, these data are relayed to the cloud using the transmission control protocol/internet protocol (TCP/IP) in conjunction with the message queuing telemetry transport (MQTT) protocol to ensure both security and privacy.[112] Moreover, it takes into consideration both complexity and accuracy of algorithms in order to achieve more precise and reliable battery prediction and diagnosis. In addition, integrating the self-response feedback function of smart materials, i.e., dynamic responsive smart batteries, which can ultimately realize efficient active control and optimal decision-making to provide high-precision and complex functions that cannot be realized by traditional BMS.

As depicted in Figure 8, it illustrates the comprehensive control framework and feasibility assessment for achieving autonomous decision control within smart batteries. During the internal state perception phase within the battery, both implanted sensors and smart materials integrated into the application are capable of detecting changes in the battery's internal information. The crucial distinction lies in how these perceived internal parameters are handled. Implanted sensors directly transmit the

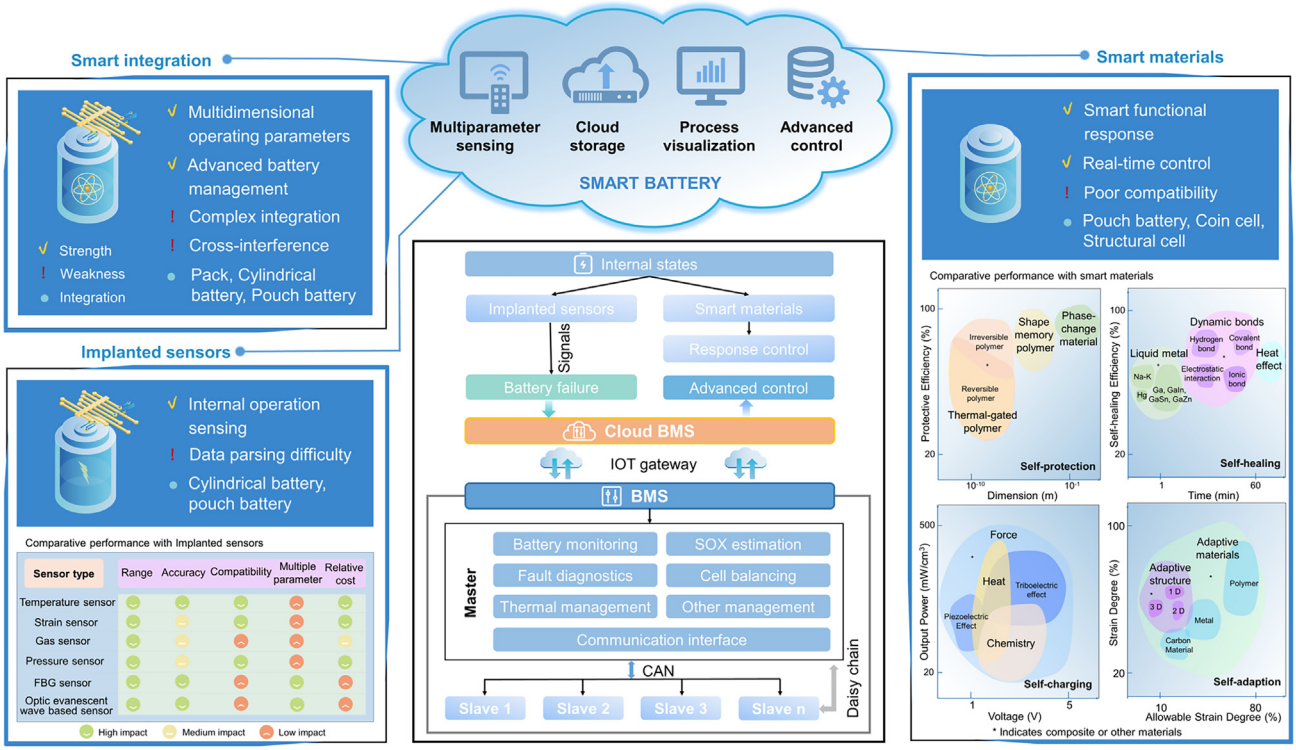


Figure 8. The comprehensive control framework and feasibility assessment of smart batteries

The decision-making process flow for smart batteries and a comparative summary of different types of sensors and the performance of various smart materials.

sensed internal parameter signals to the cloud BMS. Subsequently, the process advances through various tiers, encompassing the slave, master, and system diagnosis of battery failure to enable advanced decision control. Conversely, smart materials perceive changes in the battery's internal state and employ them as specific commands to deliver smart and timely responses and control. Within this decision-making process flow, it exemplifies a smart battery that integrates remarkable features, including multiparameter sensing, cloud storage, process visualization, and advanced control.

Although smart batteries offer numerous advantages and have promising development prospects, the feasibility of their smart integration still requires further comprehensive assessment. In Figure 8, we also have summarized the strength and weakness of implanted sensors, smart materials, and the integration of both to ultimately achieve autonomous decision-making in smart batteries.

Regarding implanted sensors, ongoing research primarily focuses on their integration into cylindrical batteries and pouch batteries. The advantages are clear because these sensors can detect the battery's internal operating state, thereby improving management performance in areas such as state estimation, charge control, safety alerts, and more. Each type of sensor has its specific strengths. Based FBG sensors, in particular, providing significant value for internal state perception. However, their commercialization is often constrained by cost considerations.[25,115,116]

In practical commercial applications, it is viable to implant various sensor types to acquire a wider array of physical and chemical parameters associated with the

battery. This approach enables the pursuit of application objectives from different angles, encompassing state estimation, lifespan assessment, and cost optimization. Nevertheless, it is crucial to acknowledge that the principal challenge in integrating multiple implanted sensors lies in data parsing difficulty. Overcoming mutual interference between various state parameters is essential to enhance perception accuracy and reinforce the reliability of monitoring and management systems.

Based on functional smart materials, batteries can be endowed with the capability for timely and smart response control. Currently, the research on battery applications primarily focuses on pouch batteries, coin cells, and structural cells. Different response mechanisms correspond to various response effects.[71,89,92,117–121] For self-protection functions, SMPs and PCMs often employ circuit blocking to achieve optimal protection. Thermal-gated polymer, on the other hand, can achieve reversible and limited protection within a specific temperature range. In the case of self-healing, mechanisms based on LM offer a relatively fast response, whereas polymers with reversible dynamic bonding have gained widespread research in battery materials. Additionally, self-charging batteries utilizing mechanical, chemical, and thermal energy can provide energy output in different voltage and power ranges, with mechanical energy utilization being more universal. The implementation of adaptive functionality primarily relies on materials, and when complemented with structured design, it can lead to an overall enhancement of battery adaptability. In summary, smart materials hold significant promise for improving battery safety and imparting functional characteristics.

However, the most substantial challenge of smart materials is the relatively poor compatibility between their functional properties and electrochemical performance. To achieve the practical application of smart materials, it is crucial to develop novel functional responsive materials, facilitate the synergistic interaction among various responsive materials, and employ strategies that combine structured design with material compatibility. These approaches are vital for enhancing the responsiveness and efficiency of smart batteries. Moreover, the response speed of self-responsive materials is a crucial factor that must meet the requirements to effectively mitigate battery failures and thermal runaway incidents. Although in scenarios where adverse conditions such as overheating are encountered, self-responsive materials can act promptly and contribute effectively to preventing thermal runaway. However, it is worth noting that their response times are relatively longer in most instances when compared with sensor-based systems (as shown in Table S2). As a result, additional research and development efforts are necessary to optimize the response speed and overall performance of these materials, ensuring they meet or exceed safety standards. Alternatively, the incorporation of embedded sensors and smart control systems can be employed to execute control commands, expediting the triggering of smart material responses.

When it comes to smart integration, the ultimate goal for smart batteries is the simultaneous incorporation of implanted sensors and smart materials within the battery. This integration aims to achieve advanced control through a cloud-based BMS. The practical battery scales suitable for such integrated designs primarily include cylindrical batteries and pouch batteries. Within this system, smart batteries not only possess the capability to perceive multi-dimensional physical and chemical information about the internal state but also gain a clearer understanding of the response mechanisms of smart materials. Beyond achieving timely response control, they can also realize higher-level autonomous decision control.

However, the design and distribution of jointly implanted sensors and smart materials present significant challenges. Their collaborative interaction may lead to cross-interference, such as the case where the response commands from smart materials correspond to the same physical parameters sensed by the sensors. This can further impact perception accuracy and response sensitivity. It can be envisioned that the future development direction will primarily concentrate on the distributed design of their combined integration, which is essential for enabling smart batteries to attain advanced autonomous decision-making capabilities. This may require a redesign of the physical structure of the battery to accommodate sensors, smart materials, and electronic components, while ensuring that the battery's performance and stability are not compromised.

OUTLOOK

To realize the development and manufacturing of smart batteries, there are still many bottlenecks to overcome in further research and development, and these challenges may bring opportunities for sustainable development. Although the research on smart batteries has become a hot topic in recent years, the exploration of their electrochemical performance, intelligent behavior, and practicality is still in its infancy. Therefore, there is a need for performance optimization and structural modification of smart batteries to form technology clusters. We will discuss the current scientific issues facing smart batteries, some technical pathways to address these challenges, and look forward to the Visio roadmap of smart batteries (Figure 9).

(1) Real-time perception smart batteries have been developed that implant multiphysical field advanced sensors inside the battery, including temperature, strain, gas, and pressure. This results in the collection and analysis of internal battery information, which further provides diagnostic information for battery management. In order to achieve accurate control of batteries by BMS, there are still problems that need to be explored and solved, including the improvement of sensor's sensing accuracy, immature analysis and integration of sensor signals with BMS, sensor signal cross-mixing phenomena, and vulnerability to electrolyte corrosion.

Based on the above, ultra-thin sensors can be manufactured and processed by high-precision processing technology in order to improve the sensing capability and produce smaller errors in the measurement range. By enhancing a single feature, the multiple sensing signal crossover phenomenon can be effectively avoided. Physically relevant features can be obtained from a large amount of data collected by advanced sensors based on an AI algorithm for comprehensive analysis, which can extract abnormal values and achieve a series of multi-dimensional and comprehensive diagnostic functions, such as battery safety warning and lifetime prognostics. Furthermore, the corrosion resistance of sensors can be improved through copolymerization and other methods, by further screening and developing corrosion-resistant encapsulation materials.

(2) The development of dynamic response smart batteries is based on applying smart materials to obtain smart functions. Smart materials play an indispensable role in the interactive response of batteries. Therefore, the development of new smart materials is essential to advance smart batteries. However, the design and development of new materials is dominated by the slow and ineffective pace of conventional experimental research models, which restricts the development of multifunctional smart batteries. To develop more advanced functions, new features are crucial for smart batteries to address the negative effects on compatibility and

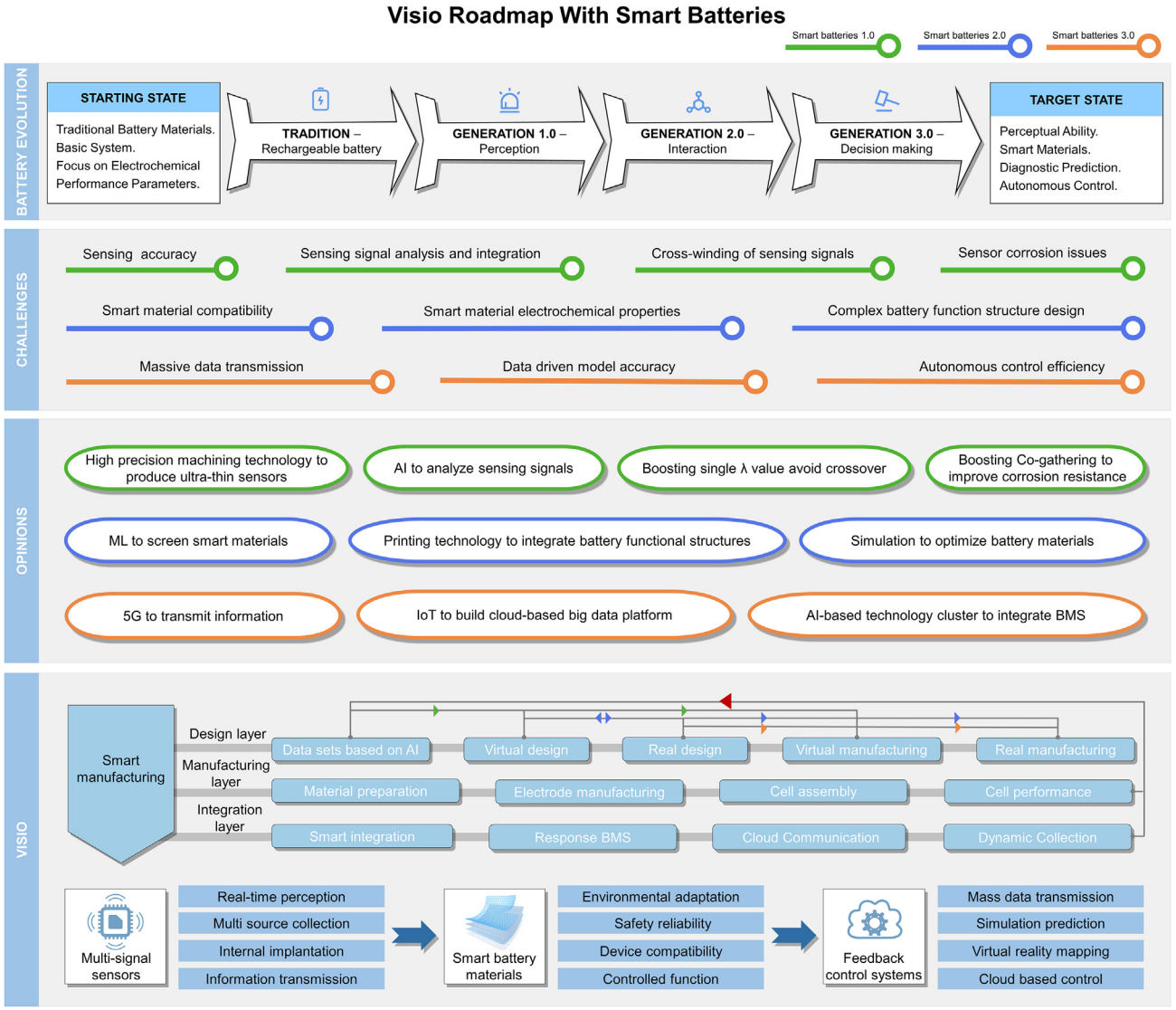


Figure 9. Visio roadmap with smart batteries

Scientific problems facing the development of smart batteries in different generations and some opinions on feasible technical ways to solve them for practical and sustainable development of smart batteries.

original electrochemical performance. Additionally, the functional complexity of the structural design of battery components poses additional challenges, resulting in smart batteries that cannot achieve maximum potential balance performance, which ultimately impacts the overall performance of the device.

The establishment of complex processing-microstructure-property relationships of materials from microscopic to mesoscale and then to macroscopic can be achieved by integrating ML and data feedstock into theoretical calculations or experiments. The method enables to predict the physical properties of materials and facilitate the design of functional materials. With AI and ML techniques, it is feasible to screen smart materials suitable for the entire smart batteries chain, thereby accelerating the design of new batteries in terms of performance, efficiency, and sustainability. Electrochemical simulations complement experiments in analyzing and optimizing

battery materials. Identifying factors that may influence the battery operation process by combining experimental and simulation data can offer an effective guide for trade-off and optimization between the overall compatibility and performance of smart materials and devices. In addition, 3D/four-dimensional (4D) printing technology has been effectively applied in preparing a variety of batteries with high shape consistency that can be used for rapid integration of complex functional structures designed in smart batteries. The most significant advantage of 3D/4D printing technology is the ability to precisely control customized shapes and structures from microscopic to macroscopic levels without relying on pre-developed templates.

(3) Self-decision-making smart battery equipped with AI, mobile communication, and automatic control technology, developed on the basis of real-time perception and dynamic response to achieve autonomous management and control of batteries. It includes battery state estimation and aging prediction, battery modeling, and management strategies to maintain performance, life, and safety during operation. However, the challenge of maintaining both the accuracy of the model and the efficiency of the control persists in the face of massive data transmission, long operation cycles, and complex interaction parameters. Therefore, it is imperative to continue researching potential solutions to these issues.

By building a cloud-based big data platform through IoT technology to integrate multiple sources of information in different time and space, combined with AI technology can greatly improve the adaptability and integration of electrochemical model architecture to achieve multi-dimensional mapping and simulation of physical batteries. Afterward, the integrated information is transmitted to the BMS via 5G technology, allowing the BMS to develop a streamlined and effective control strategy for optimizing operation.

The comprehensive development of smart batteries is inseparable from smart manufacturing. The process of battery smart manufacturing integrates advanced technologies and data analytics methods, combining virtual and real elements to achieve intelligence, efficiency, and quality assurance.[10,12] Initially, AI technology is employed to analyze and model a vast amount of data, which can be used for preliminary virtual design, enabling an accurate prediction and optimized performance. During the actual design process, key parameters such as electrode materials and electrolyte composition can be made a targeted determination. Following the actual design, virtual and real manufacturing are integrated to ensure that the battery production process is efficient, precise, and controllable. By simulating and optimizing production processes, virtual manufacturing minimizes resource wastage while validating and making adjustments in real time during actual manufacturing. Specifically, at the material level and electrode manufacturing stage, intelligent production lines ensure that electrode production complies with design specifications, while real-time adjustment capabilities ensure manufacturing flexibility. In the battery assembly process, coordination between real and virtual manufacturing ensures accurate assembly and connection of each battery component. Decoupling correlations between various manufacturing parameters and further exploring the impact of these parameters on the final battery performance are crucial aspects.

The reasonable integration technology can be regarded as a crucial step in maximizing the efficiency of smart batteries. The distributed perception and control components should be integrated with core management system. The convenience of information transmission and the connectivity of intelligent components cannot be ignored. Meanwhile, the communication between local terminals and

cloud systems also requires strong network and computability supporting. During operation, the integrated components respond dynamically to BMS, and then the control parameters and performance indicators can be recorded by the cloud system, which are used in the iterative refinement for building efficient and consistent smart batteries.

The current energy revolution focuses on promoting the intelligent transformation and upgrading of the energy industry, improving resource utilization efficiency, and accelerating the construction of a smart, efficient, clean, low-carbon, and recycling green industrial system in response to new information technologies, such as the Internet, big data, and cloud computing. Future work will prioritize the development of multifunctional high-performance smart batteries. In the long run, the construction of an autonomous energy interconnection network can be achieved through multi-signal sensors, smart battery materials, and corresponding feedback control systems combined with advanced manufacturing processes and AI-driven methodologies. In order to accelerate the growth of the smart era, the entire network design should strongly incorporate the new generation of intelligent technologies, focusing an emphasis on real-time, systematic, efficient, interactive information, as well as extensive services.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2024.01.011.

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AUTHOR CONTRIBUTIONS

Conceptualization, R.C. and Y.H.; writing – original draft, R.C., Y.H., and Q.M.; writing – review and editing, R.C., Y.H., Q.M., L.L., and F.W.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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